Radiant Hospital Management System (HMS)

ENTERPRISE ON-PREMISE SOFTWARE • TECHNICAL SPECIFICATION DOCUMENT

Deployment Model: 100% Local LAN (Air-Gapped)
Target Environment: Hospital Server (Windows 10/11/Server)
Document Release: v1.0.0 • Production Certified

Core Architectural Principle: Radiant HMS is an **independent, on-premise healthcare platform** deployed exclusively inside the hospital facility. It functions **100% offline** without reliance on external internet connections, cloud vendors, or recurring hosting subscriptions. All patient clinical records, diagnostic files, and financial ledgers remain strictly confidential within the hospital's physical walls.

Zero Internet Dependency — OFFLINE
Uninterrupted 24/7 hospital workflow during telecom disruptions or internet outages.

Complete Data Sovereignty — PRIVATE
Zero cloud fees. Full compliance with medical data localization and privacy mandates.

Zero Client Installs — WEB BROWSER
Doctors, nurses, and billing staff access via standard Chrome or Edge browsers.

Disaster Resilience — AUTOMATED
Nightly automated database backups with 10-minute full system disaster recovery.

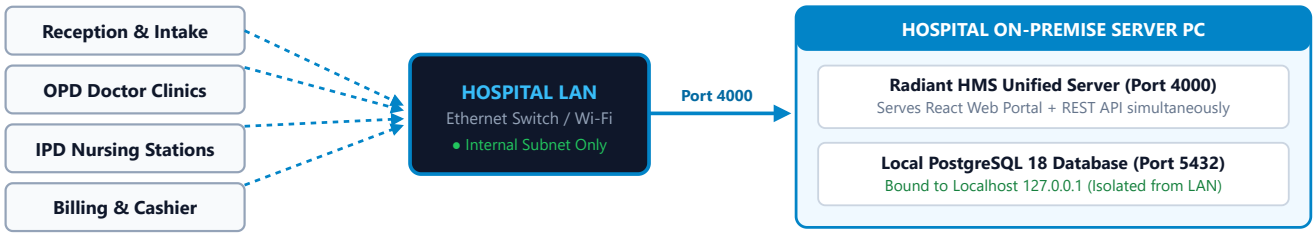
1 TECHNOLOGY STACK & ENGINEERING RATIONALE

Software Layer	Technology & Version	Clinical & Operational Benefit to the Hospital
Frontend Client UI	React 18 + TypeScript + Tailwind CSS	Ultra-fast, responsive single-page application. Zero desktop installations. Built-in pixel-perfect print formatting for invoices, prescriptions, and lab reports.
Application Server	Node.js LTS (v20+) + Express REST API	High-throughput, event-driven engine capable of handling hundreds of concurrent hospital staff requests with sub-10ms response times.
Database Engine	PostgreSQL 18 + Prisma ORM	Enterprise relational engine providing strict ACID transactional compliance, referential integrity, and automated schema migrations.
Security & Auth	bcrypt (12 rounds) + JWT Sessions	Role-Based Access Control (10 granular roles), automated session expiration, brute-force rate limiting, and immutable audit logs.

2 HOSPITAL LOCAL AREA NETWORK (LAN) PHYSICAL TOPOLOGY

Department Workstation Connectivity (Zero External Internet Exposure)

LOCAL SUBNET: 192.168.1.0/24



Modular Functional Capabilities & Security Governance

CLINICAL WORKFLOWS, ACCESS CONTROL & DISTRIBUTION PIPELINE

Security Standard: Role-Based Access Control (RBAC)
Architecture: Single-Port Embedded Distribution

3 PRE-CONFIGURED CLINICAL & ADMINISTRATIVE MODULES

Patient Management — EMR

Automated Medical Record Number (MRN) generation, duplicate patient detection and merging, complete demographic profiles, and printable registration cards.

Appointment Scheduler — QUEUE

Conflict-free doctor consultation booking, availability rosters, multi-department outpatient queues, and printable patient appointment slips.

Outpatient Department — OPD

Physician consultation notes, vital signs logging (BP, pulse, temp, weight), ICD diagnosis recording, and electronic Rx prescription builder.

Inpatient Admission — IPD

Ward and bed allocation matrix, real-time bed occupancy tracking, inter-ward patient transfers, clinical treatment plans, and discharge summaries.

Billing & Cashier — FINANCE

Service catalog, dynamic tax/GST calculation, multi-mode payment reconciliation (Cash, UPI, Card, Net Banking), and audited tax invoices.

Pharmacy & Lab — DIAGNOSTICS

Batch tracking with expiry alerts, FIFO drug dispensing, pathology test catalog, barcode sample tracking, and clinician-verified lab reports.

4 ROLE-BASED ACCESS CONTROL (RBAC) GOVERNANCE MATRIX

Staff Role	Patient Records	Appointments	Clinical (OPD/IPD)	Billing & Cashier	Pharmacy & Lab
Hospital Director / Super Admin	Full Control	Full Control	Full Access	Full Control	Full Control
Medical Doctor / Specialist	View / History	View / Schedule	Create / Edit Notes	View Only	Order Tests / Prescribe
Nursing Staff	View / Update Vitals	View Daily Queue	Bed Care / Vitals	No Access	Sample Collection
Reception / Front Desk	Intake / Register	Book / Reschedule	No Access	No Access	No Access
Billing & Accounts Officer	View Basic Info	No Access	No Access	Generate & Settle	View Billed Items
Pharmacist / Lab Technologist	View Patient MRN	No Access	No Access	Dispense / Invoice	Full Department Access

5 SINGLE-PORT EMBEDDED DISTRIBUTION ARCHITECTURE

The Distribution Package — RADIANT-HMS-V1.0.0.ZIP

The application is supplied as a self-contained, pre-compiled ZIP archive. It contains:

- **Frontend Pre-compiled Assets:** Optimized static bundle in `frontend/dist/`.
- **Backend Compiled Runtime:** Express engine in `backend/dist/`.
- **1-Click Batch Utilities:** Installation, launch, firewall, and backup scripts.

Single-Port (Port 4000) Advantage — ZERO MAINTENANCE

Unlike legacy systems requiring complex Apache or IIS configurations:

- **One Network Port:** Node.js serves both the web interface and REST APIs on Port 4000.
- **Zero Workstation Setup:** Works immediately in any browser via `http://[Server-IP]:4000`.
- **Minimal Attack Surface:** Only Port 4000 is open on the hospital server firewall.

Deployment Protocol & Disaster Recovery Specification

INSTALLATION STEPS, AUTOMATED BACKUPS & TECHNICAL FAQ

Deployment Time: Under 15 Minutes
Recovery Objective: RTO < 15 min • RPO < 24 hrs

6 SERVER HARDWARE SPECIFICATIONS & PREREQUISITES

Hardware / Environment	Minimum Specification (Up to 50 Beds)	Recommended Enterprise (100+ Beds / Concurrent Users)
Host Computer (Server)	Standard PC: Intel Core i3 / 8 GB RAM / 256 GB SSD	Dedicated PC or Tower Server: Intel Core i5/i7 / 16 GB RAM / 512 GB NVMe SSD
Operating System	Windows 10 Pro / Windows 11 Pro (64-bit)	Windows Server 2019 / 2022 (64-bit) or Windows 11 Pro
Power Resilience	600 VA UPS (Uninterruptible Power Supply)	1 kVA - 2 kVA Online UPS with power monitoring
Software Dependencies	1. **Node.js LTS (v20+)** (Free installer from nodejs.org) • 2. **PostgreSQL 18** (Free installer from postgresql.org)	

7 THREE-STEP AUTOMATED WINDOWS DEPLOYMENT PROTOCOL

STEP 1
Extract Archive
Extract `radiant-hms-v1.0.0.zip` directly into `C:\RadiantHMS\` on the designated server PC.

STEP 2
1-Click Initialization
Right-click `scripts\install-windows.bat` → *Run as Administrator*. Automatically initializes all 32 relational tables.

STEP 3
Firewall & Launch
Run `setup-firewall.bat` once, then double-click `start-hms.bat`. System launches and displays the Setup Wizard at `localhost:4000`.

8 AUTOMATED BACKUP & 10-MINUTE DISASTER RECOVERY POLICY

Automated Nightly Backups (2:00 AM) SCHEDULED

- Windows Task Scheduler runs `scripts\backup-db.bat` every night automatically.
- Creates a timestamped, compressed `.sql` archive in `C:\RadiantHMS\backups\`.
- Built-in 30-day retention automatically removes obsolete files to prevent disk overflow.
- **Best Practice:** Hospital IT attaches a secondary external USB drive for weekly offsite mirroring.

10-Minute Emergency System Recovery DISASTER READY

- If the server PC ever experiences total hardware failure:
- Copy the latest backup snapshot to any spare Windows laptop or PC.
- Run `scripts\restore-db.bat`, specify the backup file, and type `RESTORE`.
- All patient charts, active ward beds, invoices, and audit histories are restored with zero data corruption.

9 HOSPITAL IT & MANAGEMENT FREQUENTLY ASKED QUESTIONS

Q: What happens if our hospital's internet connection goes down?
Operations continue completely uninterrupted. Radiant HMS does not require external cloud servers, third-party CDNs, or internet access to function. All records, billing, and registrations occur over the hospital's local network.

Q: Do we need to install software on doctors' laptops, reception desks, or tablets?
No software is installed on client workstations. Staff simply open their web browser (Google Chrome, Microsoft Edge, Safari) and navigate to the hospital server's local address (e.g., `http://192.168.1.11:4000`).

Q: How does the system ensure patient data privacy and prevent leaks?
The database is bound exclusively to the server's internal localhost and is never directly accessible over the network. Passwords use 12-round salted hashing, permissions are strictly checked by server middleware, and an immutable audit log records all sensitive modifications.

Q: Can we connect mobile apps or medical diagnostic machines in the future?
Yes. Radiant HMS is built around standard RESTful JSON APIs and Swagger documentation. Mobile applications (iOS/Android) and laboratory equipment can integrate directly with existing endpoints without changing the backend.

Certified Release: Radiant HMS Production v1.0.0 • 100% On-Premise Verified

Technical Support & Warranty: Full Installation Assistance, Schema Migrations & Documentation Included